Experiment 10

Title **ElderWatch – Smart Elder Safety & Health Monitoring System**

Mini Project

Abstract:

Objectives:

To design a dynamic web application using front-end and back-end technologies

To implement CRUD operations using PHP and MySQL

To retrieve data from a public API using AJAX / Fetch API

To implement dynamic search and filtering

To ensure secure data handling using validation, sessions, and cookies

To understand real-world client–server interaction

Problem Statement:

Elderly people living alone face serious challenges:

- No daily health monitoring
- Delayed emergency response No immediate alert system Forgetting medicines
- Family unable to track location during emergencies

There is a need for a simple, accessible, and
affordable monitoring system.

Proposed system

ElderWatch provides:

- Daily Check-In System
- Emergency SOS Button
- Medicine Reminder & Tracker
- Emergency Contact Network
- Location Tracking
- Voice Command Support
- Family Notification System

It ensures safety, faster response, and better monitoring.

software /hardware requirements

Hardware Requirements	Software Requirements
Laptop / Computer	HTML, CSS, JavaScript
Processor: Intel i3 or above	Python (Flask)
RAM: Minimum 4 GB (8 GB recommended)	SQLite / MySQL
Storage: 10–20 GB free space	Web Speech API
Smartphone	Geolocation API
Internet Connection	Twilio (SMS), Email SMTP
GPS-enabled device	Manual Testing
Microphone & Speaker	GitHub
	VS Code

conclusion

The Multifunctional Smart Healthcare System and Automated Alert Mechanism was developed successfully to improve the safety of elderly users by providing features like SOS alerts, medication reminders, emergency contact management, and location tracking. The system ensures quick communication during emergencies through SMS, email, and WhatsApp. The project achieved its objective of creating a simple, reliable, and user-friendly healthcare support system.

Chapter 1

Introduction

1.1. Problem Statement

In today's fast-paced world, ensuring the safety and healthcare of elderly individuals living alone is a major challenge. Existing systems mainly monitor basic health parameters but fail to provide immediate assistance during emergencies, leading to delays and serious consequences. Elderly people often struggle to seek help due to limited mobility, lack of nearby support, and communication difficulties. Additionally, there is no integrated system that combines emergency alerts, medication tracking, real-time monitoring, and communication in one platform. Most solutions rely on a single communication method and are often complex for elderly users. Therefore, a simple, reliable, and multifunctional system is needed to provide quick emergency response, continuous monitoring, and effective communication, ensuring safety and supporting independent living.

1.2. Aim and Objectives

Aim:

To develop a smart healthcare system for elderly users that ensures safety and health management by providing quick emergency assistance and integrating alerts, monitoring, and communication in a simple platform. [4].

Objectives:

The primary objectives of the Elderwatch project are systematically defined as follows:

- To develop a smart healthcare system that ensures the safety of elderly individuals living independently.

- To provide a quick and reliable emergency SOS alert system with real-time location sharing.

- To implement a daily health check-in feature to monitor user well-being.

- To design a medication tracking system with reminders and status updates.

- To enable real-time location monitoring for caregivers using GPS.

- To integrate multi-channel communication (SMS, WhatsApp, Email) for effective alert delivery.

- To create a simple and user-friendly interface suitable for elderly users.

- To support voice-based interaction for hands-free hands-free operation.

1.3. Project Scope

The scope of the ElderWatch system is to provide a smart, web-based healthcare and safety solution for elderly individuals living independently. The system focuses on monitoring user well-being and ensuring quick assistance during emergency situations.

It includes features such as an emergency SOS alert system with real-time location sharing, daily health check-ins, medication tracking, and live location monitoring. The system enables caregivers or family members to receive instant alerts through multiple communication channels, including SMS, WhatsApp, and Email.

ElderWatch is designed with a simple and user-friendly interface to ensure ease of use for elderly users. It operates through a browser-based platform using GPS-enabled devices, making it accessible without the need for complex installations

The project is limited to a prototype implementation using a Flask backend and SQLite database, suitable for small-scale usage. Future enhancements may include mobile application development, integration with wearable health devices, cloud deployment, and advanced health analytics.

Chapter 2

Literature Review

2.1 Survey of Existing System/SRS

In recent years, several healthcare and monitoring systems have been developed to support elderly individuals. Existing solutions such as wearable health trackers and mobile health applications mainly focus on monitoring basic parameters like heart rate, steps, and sleep patterns. Some systems also provide emergency alert features; however, they are often limited to a single communication channel such as SMS or app notifications.

Many available systems lack integration of multiple essential features like real-time location tracking, medication management, and emergency response within a single platform. Additionally, most systems are dependent on wearable devices, which may not always be convenient or affordable for all users.

Another limitation of existing systems is their complexity. Many applications are not designed with elderly users in mind, making them difficult to operate due to complicated interfaces and navigation. Furthermore, delays in communication and lack of multi-channel alert mechanisms reduce the effectiveness of these systems during critical situations.

Therefore, there is a need for a simple, integrated, and reliable system that combines emergency alerts, health monitoring, and communication features in one platform. The proposed ElderWatch system aims to address these gaps by providing a user-friendly interface, real-time monitoring, and multi-channel alert support for improved safety and efficiency.[3]

2.2 Limitation: Existing system or Research gap

- Most existing systems focus only on basic health monitoring and do not provide instant emergency assistance.
- Many solutions rely on a single communication channel (e.g., only SMS or app notification), which may fail during critical situations.
- Lack of integration—features like SOS alerts, medication tracking, real-time location, and communication are not available in one unified platform.
- Many systems depend on wearable devices, which may not be convenient or affordable for all users.
- Userinterfaces are often complex and not elderly-friendly, making them difficult to operate. • Delays in alert delivery and response reduce effectiveness during emergencies.
- Limited support for real-time tracking and continuous monitoring.[6]

2.3 Mini Project Contribution

- Designed and developed a web-based smart healthcare system for elderly safety and monitoring.
- Implemented a multi-channel emergency SOS system that sends alerts via SMS, WhatsApp, and Email with live location.
- Developed a daily health check-in feature to monitor user well-being.
- Created a medication tracking module with reminders and status logging. Integrated real-time location tracking using the browser's Geolocation API and map visualization.
- Built a voice assistant feature for hands-free operation using speech recognition.
- Structured the system using a Flask backend, SQLite database, and frontend with HTML, CSS, and JavaScript.
- Ensured a simple and user-friendly interface suitable for elderly users..[2]

Nehali vaze– Integration & Features Implementation

- IntegratedAPIs like Twilio (SMS/WhatsApp) and SendGrid (Email)
- Implemented real-time location tracking using GeolocationAPI

- Workedonvoice assistant using Web SpeechAPI
- Tested system functionality and handled debugging

Priti Yadav: Backend & Database Management

- Developed backend using Flask framework
- CreatedAPI routes (SOS, voice command, data handling)
- Designed database using SQLite and SQLAlchemy
- Manageddata storage for users, contacts, medications, and logs

Aishwarya Marchande: Frontend & UI Development

- Designed and developed user interfaces using HTML, CSS, and JavaScript
- Implemented dashboard, medication page, and layout structure
- Handled user interactions like buttons, forms, and navigation
- Workedonresponsive and user-friendly design for elderly users

Chapter 3

Proposed System

3.1 Introduction

The proposed system, ElderWatch, is a smart healthcare and safety monitoring solution designed specifically for elderly individuals living independently. It aims to overcome the limitations of existing systems by providing an integrated platform that combines emergency response, health monitoring, and communication features in a simple and user-friendly interface.

The system focuses on ensuring user safety by providing a quick and reliable emergency SOS mechanism that sends real-time alerts along with the user's location to caregivers through multiple communication channels such as SMS, WhatsApp, and Email. In addition, it includes features like daily health check-ins, medication tracking, and real-time location monitoring to support continuous supervision of the user's well-being.

ElderWatch is developed using a web-based client-server architecture, making it accessible through standard devices without requiring complex installations. The system is designed with simplicity in mind, ensuring that elderly users can operate it easily with minimal technical knowledge. By integrating multiple essential functionalities into a single platform, the proposed system enhances safety, reduces response time during emergencies, and promotes independent living.

3.2. Architecture/ Framework/Block diagram

1. User Interface: Features like login, SOS button, "I'm OK," reminders, and contacts
2. Authentication Module: Handles user registration, login, and security
3. Application Layer: Core functions like SOS alerts, status checks, and contact handling
4. BackendServer: Processes requests and triggers alerts
5. Database: Stores user data, contacts, medication info, and alert history
6. Notification Module: Sends alerts via SMS, email, and WhatsApp

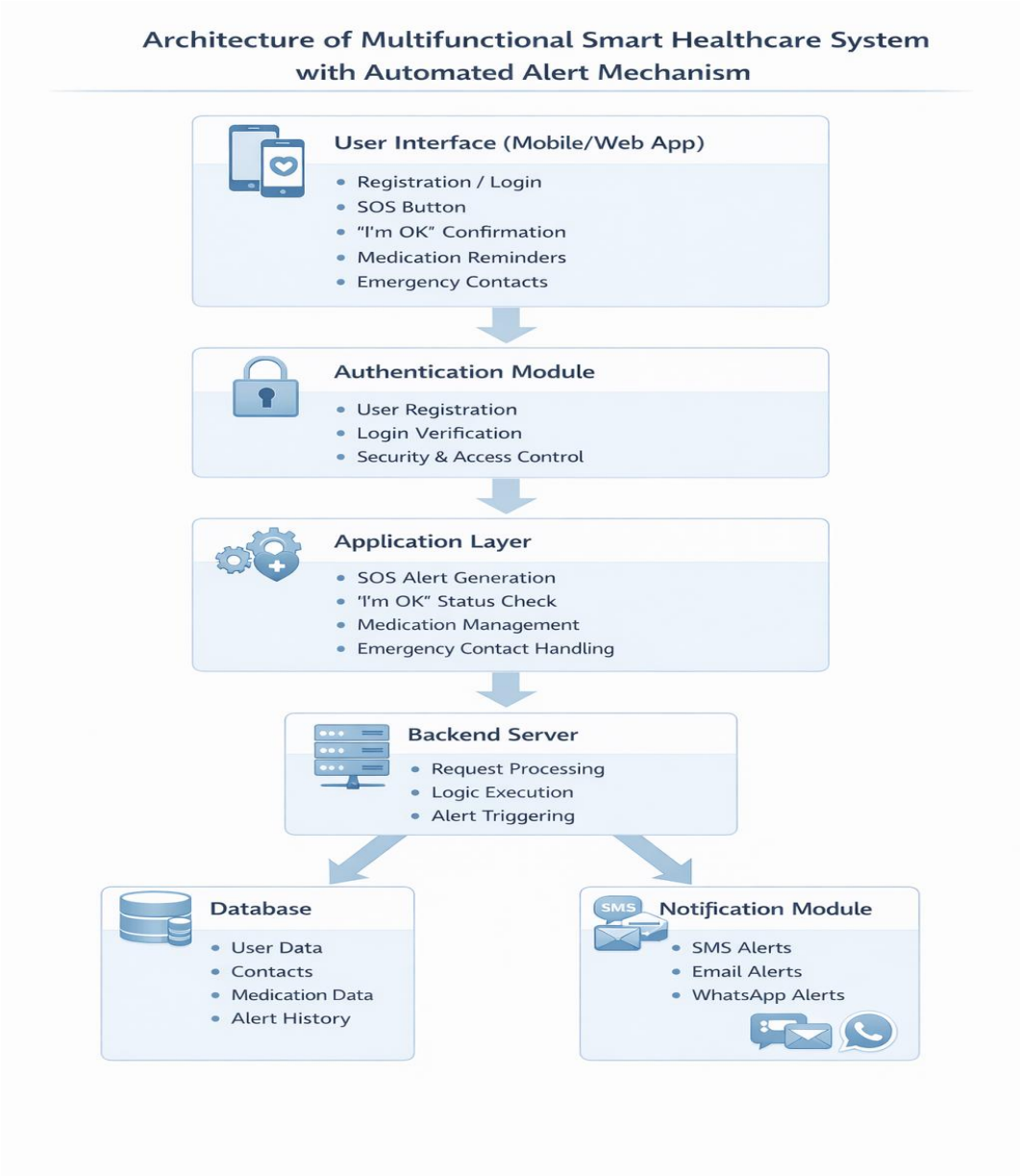

Fig 3.2.1

3.3 Algorithm and Process Design

Algorithm

1. Start
2. User clicks SOS button
3. Capture user's location using Geolocation API
4. Send data to backend server
5. Store event in database

6. Fetch emergency contacts
7. Send alerts via SMS, WhatsApp, and Email
8. Display confirmation message
9. End

Process Design

The system follows a client-server process where the frontend collects user input and sends it to the backend. The backend processes the request, interacts with the database, and triggers external APIs for communication. The response is then sent back to the frontend for user notification.

3.4 Details of Hardware & Software

To ensure optimal performance and scalability, the elderwatch platform uses a lightweight yet powerful hardware and software setup.
Client-Side Hardware Requirements:
Elderwatch is designed for high accessibility with minimal requirements:

Hardware Requirements

- Computer/Laptop with minimum 4 GB RAM
- Processor: Intel i3 or higher
- Internet connectivity
- GPS-enabled device (for location tracking

Software Requirements

- Operating System: Windows/Linux/MacOS
- Programming Language: Python
- Framework: Flask
- Database: SQLite (with SQLAlchemy ORM)
- Frontend: HTML, CSS, JavaScript
- APIs:Twilio (SMS/WhatsApp), SendGrid (Email)

- Browser: Google Chrome / any modern web browser

3.5 Experiment and Results for Validation and Verification

To verify the proper working of the **Multifunctional Smart Healthcare System and Automated Alert Mechanism**, different modules of the system were tested under various conditions. The objective was to validate whether the system performs the required functions correctly and reliably.

1. User Registration and Login Testing

The registration and login module was tested using valid and invalid user credentials.

Result:

The system successfully allowed valid users to register and log in, while invalid login attempts were rejected. This verified the secure authentication functionality.

2. SOS Emergency Alert Testing

The SOS emergency button was tested to check whether alerts were sent to emergency contacts.

Result:

When the SOS button was pressed, emergency alerts were successfully delivered to the saved contacts through SMS, email, and WhatsApp. This verified the emergency alert mechanism.

3. Medication Reminder Testing

The medication reminder feature was tested by setting reminder times.

Result:

The system generated reminders at the correct scheduled time, confirming that the reminder module worked properly.

4. Emergency Contact Testing

Emergency contact details were added and retrieved from the database.

Result:

The contact information was stored and accessed successfully, confirming proper database functionality.

5. Location Tracking Testing

The location tracking feature was tested during emergency alert generation.

Result:

The system successfully fetched and shared the user's real-time location with emergency contacts, verifying location tracking capability.

Overall Validation Result

All modules were tested successfully, and the system performed as expected. The results confirm that the proposed system provides reliable emergency support, timely reminders, and efficient communication for elderly healthcare monitoring.

Result images:

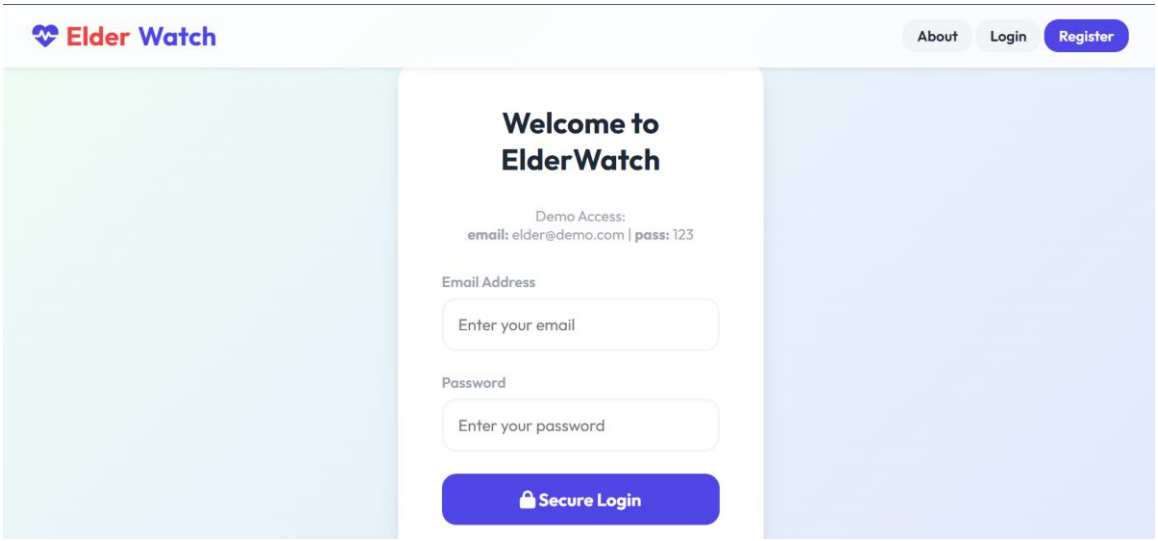

Fig 3.5.1 login page

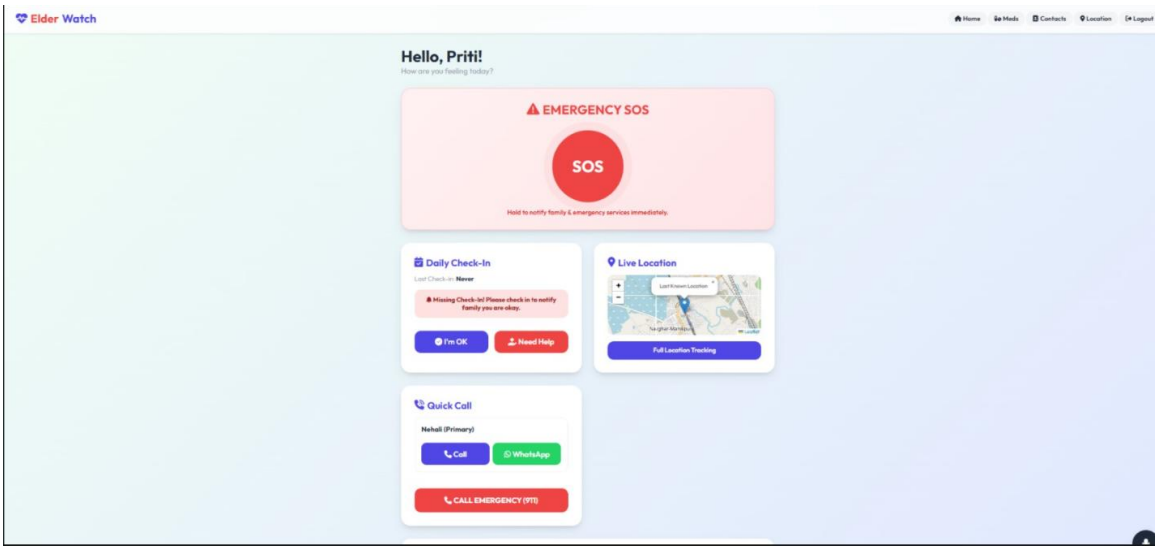

fig 3.5.2 Dashboard page

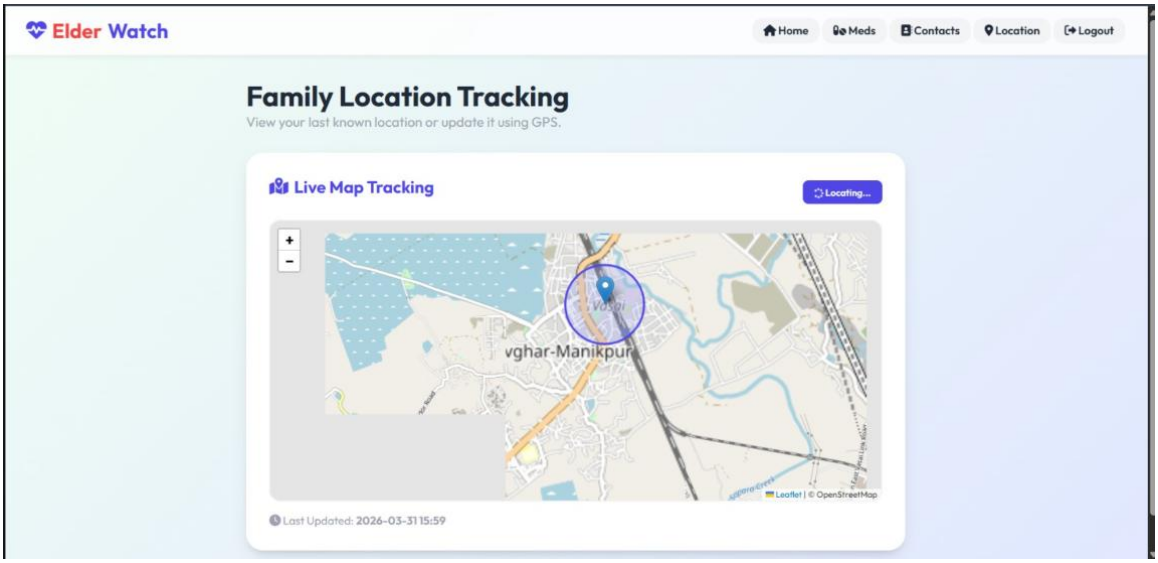

Fig 3.5.3 Family Location Tracking

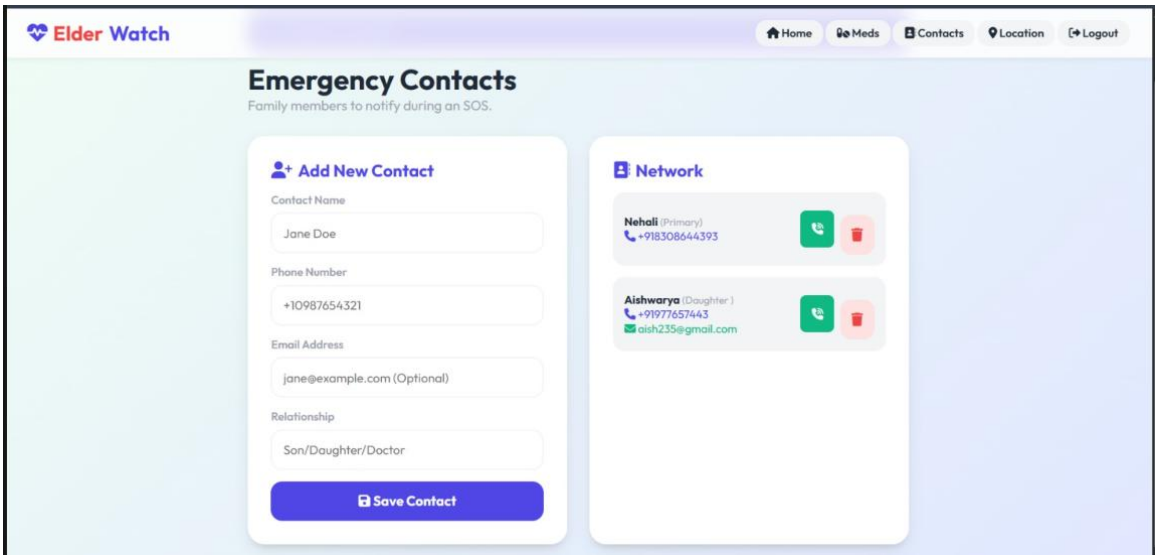

Fig 3.5.4 Emergency Contacts

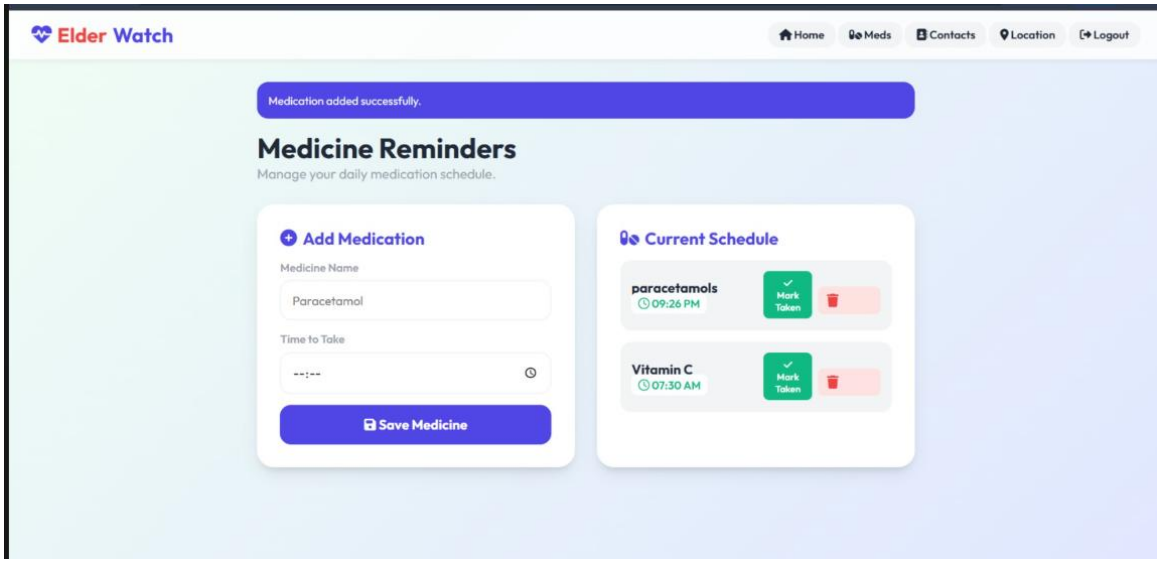

Fig 3.5.5 Medicine Reminder

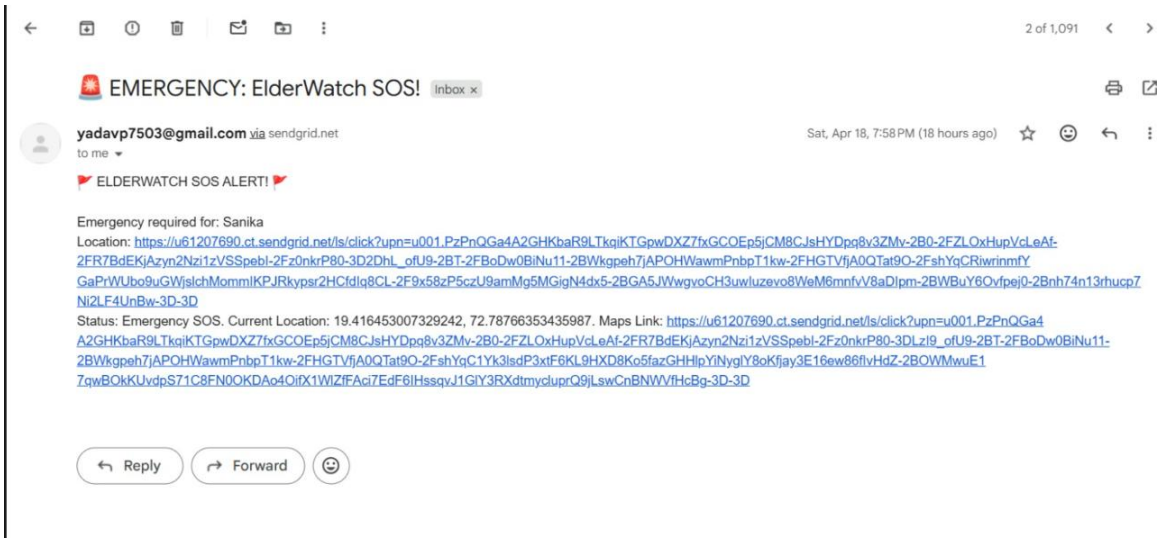

Fig 3.5.7 E-Mail Alert

3.6 Analysis

The ElderWatch system was analyzed based on functionality, usability, and reliability. The system successfully performs key operations such as emergency alert generation, real-time location tracking, medication management, and daily health check-ins. The integration of multiple communication channels ensures that alerts are delivered quickly and reliably during emergencies.

From a usability perspective, the interface is simple and easy to navigate, making it suitable for elderly users. The voice assistant further improves accessibility by enabling hands-free interaction

The system demonstrates efficient performance for small-scale usage using a Flask backend and SQLite database. However, scalability may be limited for large-scale deployment, which can be improved by using more advanced databases and cloud infrastructure.

Overall, the system effectively meets its objectives by providing a reliable, integrated, and user friendly healthcare monitoring solution.

3.7 Conclusion and Future work

Conclusion:

The ElderWatch system successfully provides a smart and reliable solution for monitoring the safety and health of elderly individuals living independently. By integrating features such as emergency SOS alerts, real-time location tracking, medication management, and daily health check-ins, the system ensures quick response during critical situations.

The use of a simple and user-friendly interface makes it accessible for elderly users, while multi channel communication enhances reliability. Overall, the project achieves its objective of improving safety, reducing response time, and supporting independent living.

Future Work:

The ElderWatch system can be further enhanced by adding advanced features and improving scalability. A dedicated mobile application (Android/iOS) can be developed for better accessibility and real-time notifications. Integration with wearable devices such as smartwatches can enable automatic health monitoring and fall detection.

The system can be deployed on cloud platforms with a more scalable database like MySQL or PostgreSQL to support large-scale usage. Advanced features like AI-based health prediction, anomaly detection, and personalized alerts can also be implemented.

Additionally, improvements in data security, user authentication, and real-time push notifications can make the system more robust and reliable. These enhancements will further improve the effectiveness and usability of the system in real-world applications

References

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Chapter 4

Annexure

4.1 Proof of Concept

The **Multifunctional Smart Healthcare System and Automated Alert Mechanism** was developed as a proof of concept to demonstrate the feasibility of providing emergency healthcare support for elderly individuals through an integrated digital platform.
The system includes the following functional modules:

- **User Authentication Module** for secure registration and login
- **SOS Emergency Alert Module** for sending emergency notifications
- **Medication Reminder Module** for timely medicine alerts
- **Emergency Contact Management Module** for storing contact details
- **Real-Time Location Tracking Module** for locating the user during emergencies
- **Multi-Channel Notification Module** for sending alerts through SMS, Email, and WhatsApp

The successful execution of these modules confirms that the proposed system can effectively improve emergency response and healthcare assistance for elderly users.
Screenshots of the developed system are attached below as proof of concept.

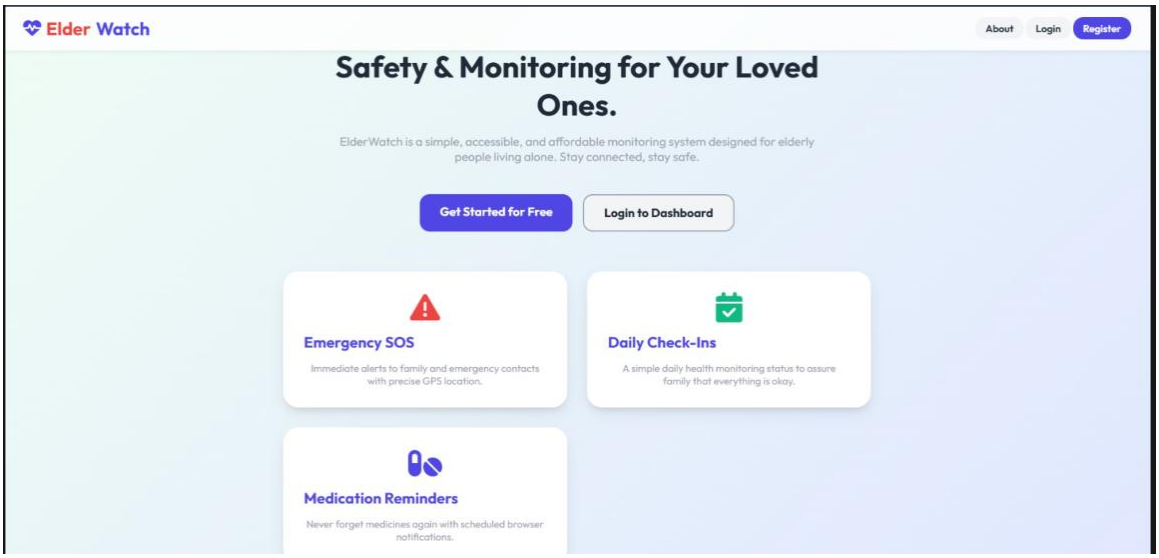
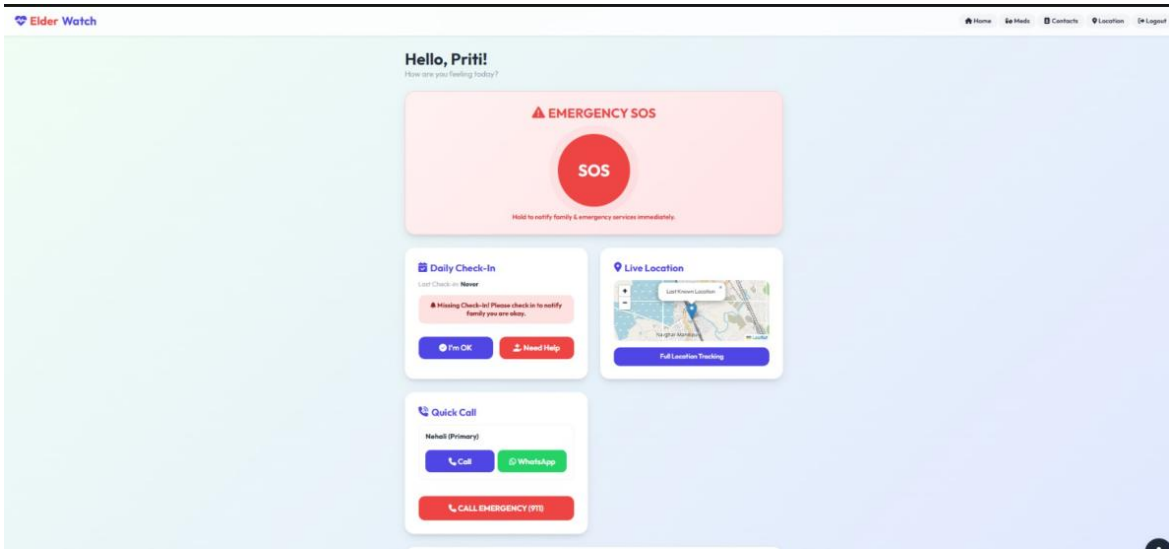


Fig. 4.1.A. Landing Page of Elderly Monitoring Application



4.1B Dashboard